| Surname and Initials: | Session Day: | Group no |
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| Reyneke, R | Wed 6 | 6 |
| Student number: | | |

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

# DEPARTMENT OF PHYSICS
# FSK 116
# Practical Test
# 2021/06/21

**Total marks: 50**

1. **PART A** is to be answered on the answer sheet given. (In the case of calculations, you can answer on a separate sheet of lined paper, but clearly mark your answers on the sheet, and indicate where to find your answer in the test document).
2. Graphs are to be drawn on the **graph paper provided** (DO NOT use excel or any other graphing software, or normal lined/plain paper.)
3. **PART B** (Multiple Choice) will be answered on ClickUP.
4. Each Multiple Choice question has only **ONE CORRECT** answer.
5. Each question counts two marks.
6. Pay attention to significant figures.
7. **Declaration:** The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

| Question | Out of | Marks |
|---|---|---|
| Part A | 20 | |
| Part B | 30 | |
| Total | 50 | |

**[20]**

## PART A:
## SIMPLE HARMONIC MOTION

**Given:**

Period in a pure SHM system $T = 2\pi\sqrt{\frac{m}{k}}$ **Equation (1)**

Period of a simple pendulum $T = 2\pi\sqrt{\frac{l}{g}}$ **Equation (2)**

A mass was tied to the end of a string. The other end of the string was secured to a stand, forming a simple pendulum. The mass was displaced 7cm from equilibrium position, and then released to swing back and forth. The time for the pendulum to make 10 oscillations was measured and recorded in a table. This time was taken 5 times. The procedure was repeated for 6 different lengths of the string.

$l_1 = 140\ cm$ $l_3 = 90\ cm$ $l_5 = 40\ cm$
$l_2 = 115\ cm$ $l_4 = 65\ cm$ $l_6 = 15\ cm$

**(a)** The table of results is shown below. Complete the following table and calculate the average period ($T_{ave}$) values for each length. *(Remember to use SI units)* [4 Marks]

Title: Table to show how the lenth of the pendulum, l (m), infuences the Period, T(s).

| Measurement: | 1st length: $l_1$ = 1,4 m | | 2nd Length: $l_2$ = 1,15 m | | 3rd Length: $l_3$ = 0,9 m | |
|---|---|---|---|---|---|---|
| | Time for 10 oscillations | Period (T) (s) | Time for 10 oscillations | Period (T) (s) | Time for 10 oscillations | Period (T) (s) |
| 1 | 23.49 | 2.349 | 22.33 | 2.233 | 19.87 | 1.987 |
| 2 | 24.27 | 2.427 | 22.47 | 2.247 | 20.04 | 2.004 |
| 3 | 23.02 | 2.302 | 21.98 | 2.198 | 20.18 | 2.018 |
| 4 | 23.62 | 2.362 | 22.04 | 2.204 | 19.98 | 1.998 |
| 5 | 23.55 | 2.355 | 21.54 | 2.154 | 20.06 | 2.006 |
| **Average Period** | | 2,359 | | 2,207 | | 2,003 |

| Measurement: | 4th length: $l_4$ = 0,65m | | 5th Length: $l_5$ = 0,4m | | 6th Length: $l_6$ = 0,15m | |
|---|---|---|---|---|---|---|
| | Time for 10 oscillations | Period (T) (s) | Time for 10 oscillations | Period (T) (s) | Time for 10 oscillations | Period (T) (s) |
| 1 | 16.87 | 1.687 | 13.06 | 1.306 | 7.930 | 0.7930 |
| 2 | 17.38 | 1.738 | 12.76 | 1.276 | 7.330 | 0.7330 |
| 3 | 17.08 | 1.708 | 12.92 | 1.292 | 6.970 | 0.6970 |
| 4 | 16.98 | 1.698 | 13.22 | 1.322 | 8.000 | 0.8000 |
| 5 | 16.90 | 1.690 | 13.14 | 1.314 | 7.230 | 0.7230 |
| **Average Period** | | 1,704 | | 1,302 | | 0,7492 |

**(b)** What do you notice about the periods of the different lengths of pendulums? [1 Mark]

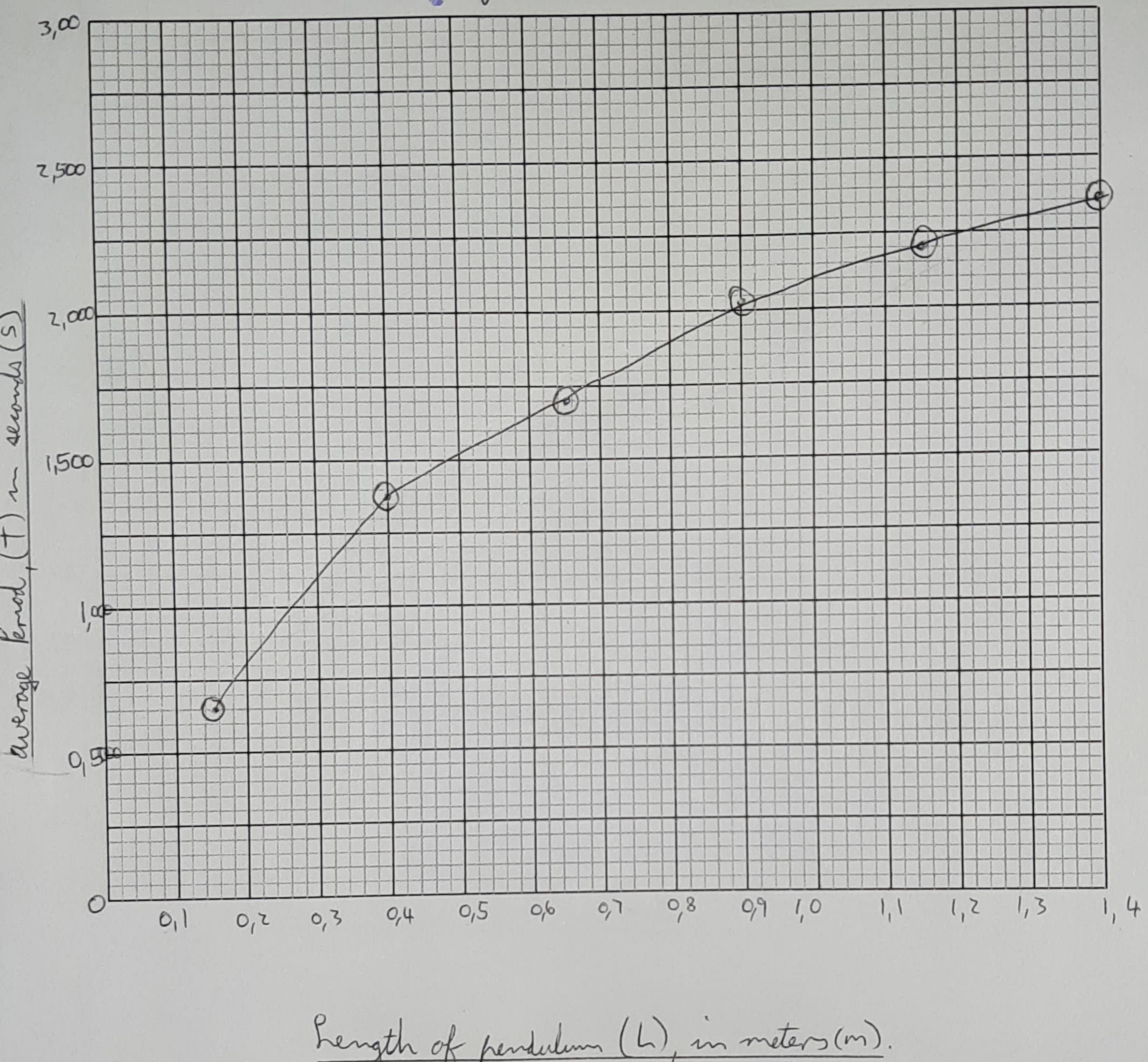
The shorter the length of the pendulum, the shorter the period. Period is directly proportional to the length of the pendulum.

**(c)** Using the average periods in Table 1, plot a graph of *T vs L*. *(Remember to apply the lab graph rules properly).* [2 Marks]

**(d)** Using the same data, plot a graph of $T^2$ *vs L*. *(Remember to apply the lab graph rules properly).* [4 Marks]

**GRAPH**

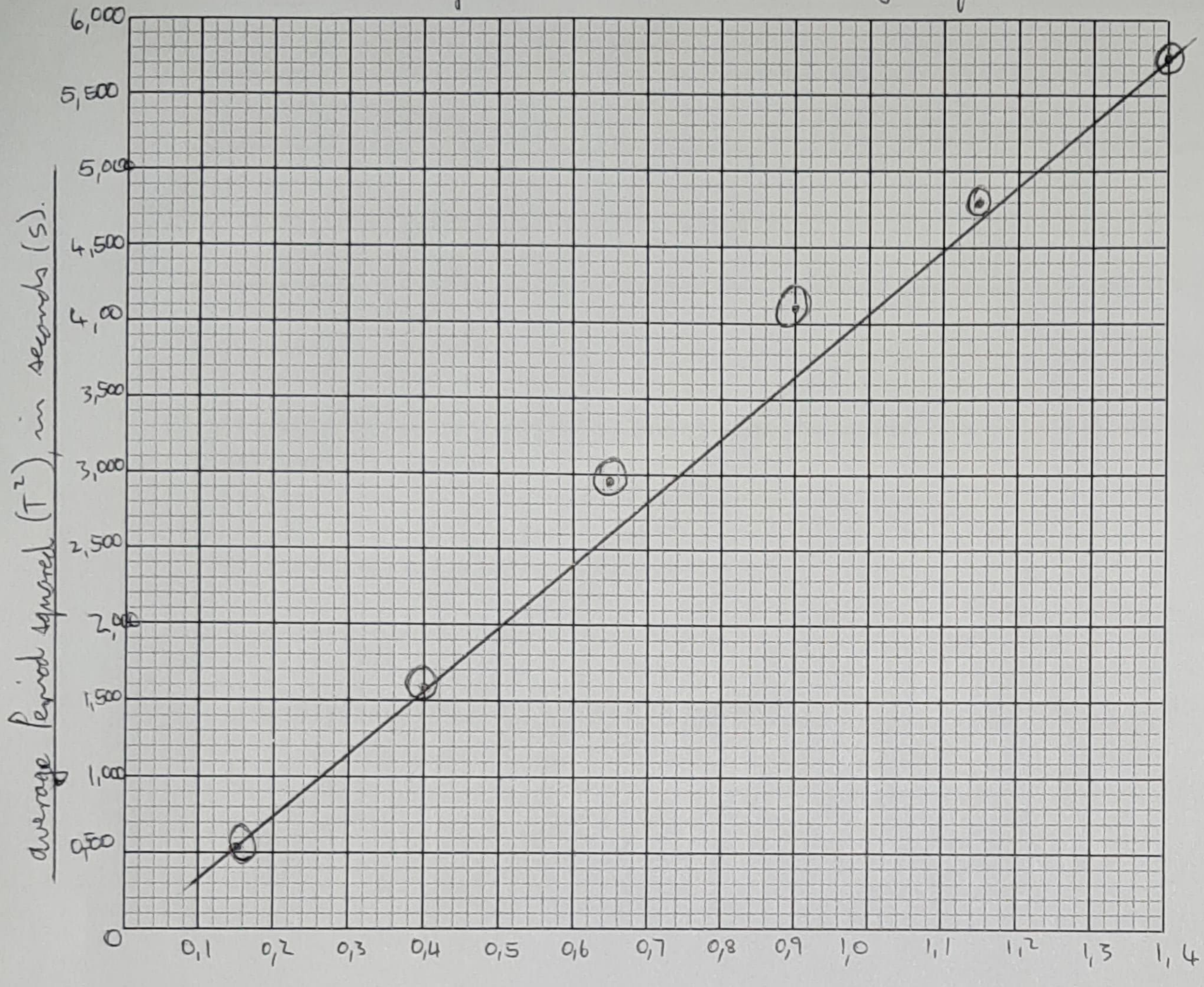
Graph to show the relation between average Period, (T) in seconds (s), and Lenth of pendulum, (L) in meters (m).




**NOTE**: *Apply the lab* ***graph rules*** *properly to earn full marks!*

**GRAPH 2** *($T^2$ vs L)*

Graph to show the relation between the average Period squared ($T^2$) in seconds (s), and the Length of a pendulum (L), in meters (m).




Lenth of pendulum (L), in meters (m).

**NOTE**: *Apply the lab* ***graph rules*** *properly to earn full marks!*

**(e)** Determine the **slope** of the second graph *($T^2$ vs L)*: [2 Marks]

$m = \frac{\Delta y}{\Delta x}$

$= \frac{(5{,}565 - 0{,}5613)}{(2{,}359 - 0{,}7492)}$

$= 3{,}108$

**(f)** Write down the empirical relationship (the equation of your line) with values and units for the constants in the equation. [2 Marks]

$T^2 = 3{,}108\,L + 1{,}215$ where $T^2$ = Period ($s^2$); $L$ = length (m); $m = 3{,}108$ ($s^2/m$)

**(g)** Using equation (2), what is the expression for T-squared? [1 Mark]

if $T = 2\pi\sqrt{\frac{\ell}{g}}$

then $T^2 = \left(2\pi\sqrt{\frac{\ell}{g}}\right)^2$

$= \frac{4\pi^2 \ell}{g}$

**(h)** Compare your empirical equation in question (f) with the mathematical expression you got in question (g). What is the mathematical relationship between the slope of your graph and *g (gravitational acceleration)*, according to your comparison? *(Provide symbols)* [2 Marks]

g is indirectly proportional to the slope of the graph

$g \propto \frac{1}{\text{slope}} = g \propto \frac{1}{m}$

$\therefore$ If g increases, the slope will decrease.

**(i)** Determine the value of $g$, the gravitational acceleration, using your answer in question (h).

[2 Marks]

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}}$$

$$\therefore g = 4\pi^2 \cdot \frac{L}{T^2}$$

$$g = 4\pi^2 \cdot \frac{(1,4)}{(2,359)^2}$$

$$= 9,93 \ m/s^2$$

# END